California State University, Fresno
Lyles College of Engineering
Electrical and Computer Engineering Department

TECHNICAL REPORT

Experiment Title: DC Shunt Wound Generators

Course Title: ECE 121L Principles of Electrical Circuits Laboratory

Date Submitted: 3/21/2024

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INSTRUCTOR SECTION

Comments: ______________________________

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Final Grade: Roberto Carlos Barajas: ______________

Joseph Igot: ______________

Rafael Lopez: ______________

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1. STATEMENT OF OBJECTIVES

Understand the behavior of generator voltage under varying rpm, the impact of excitor current changes, and the effects of altering the polarity and load on the generator's output.

2. THEORETICAL BACKGROUND

DC Shunt-wound generators are a type of DC generator where the field windings are connected in parallel (or shunt) to the armature winding.

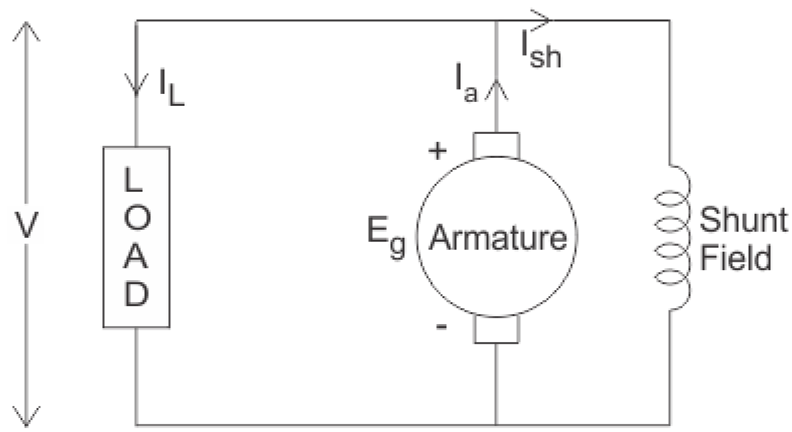


Figure 2.1 DC Shunt Wound Generator

The armature supplies an initial voltage (Eg), and this causes a current to flow through the shunt field. Current flowing through the shunt field (exciter current, or Ish), creates a magnetic field. This magnetic field increases the more exciter current there is. Increasing the rotational speed (RPM) of its armature in the magnetic field results in the induced voltage from the armature (Eg) to increase because of Faraday's Law. This feedback loop continues until the magnetic field stabilizes (when increasing the shunt field current will stop increasing the magnetic field). This plus the shunt field being parallel to the load results in the load having a stable voltage being supplied to it no matter what the load is.

Ideally, the output voltage should be the same no matter what the load is, but it can vary because when increasing the load slightly decreases the speed, and this causes the output voltage to decrease. Load characteristics of a generator will show how much of an impact the load will have on the output voltage.

The polarity of the generator's output voltage is determined by the direction of the magnetic field and the direction of rotation of the armature, and this means that reversing the polarity of the exciter winding or the rotation direction of the motor will invert the generator's output voltage.

3. EXPERIMENTAL PROCEDURE

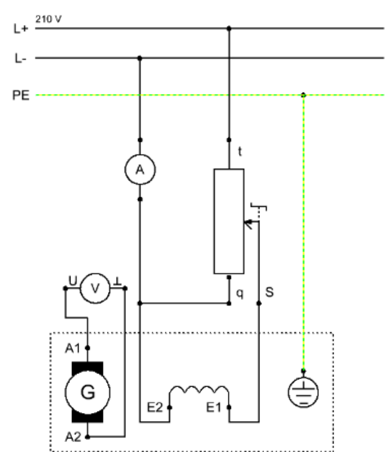
3.1. Equipment Used

- **Electrical Wires**
- **DC Machine 300 W Industrial Seres**
- **Load resistor for generator experiments**
- **Field regulator for DC Machines**
- **Coupling guard 300 W**
- **Coupling sleeve 300 W**
- **Power supply for electrical machines**
- **Analog/digital multimeter, wattmeter and power factor meter.**

3.2. Procedure

Part 1

1. **We began by connecting the DC Shunt-wound generators, separately excited. We followed the figures below to wire.**



Circuit diagram for DC shunt-wound generator, separately excited
"Voltage control"

Figure 3.2.1 DC Shunt-wound generators, separately excited circuit diagram

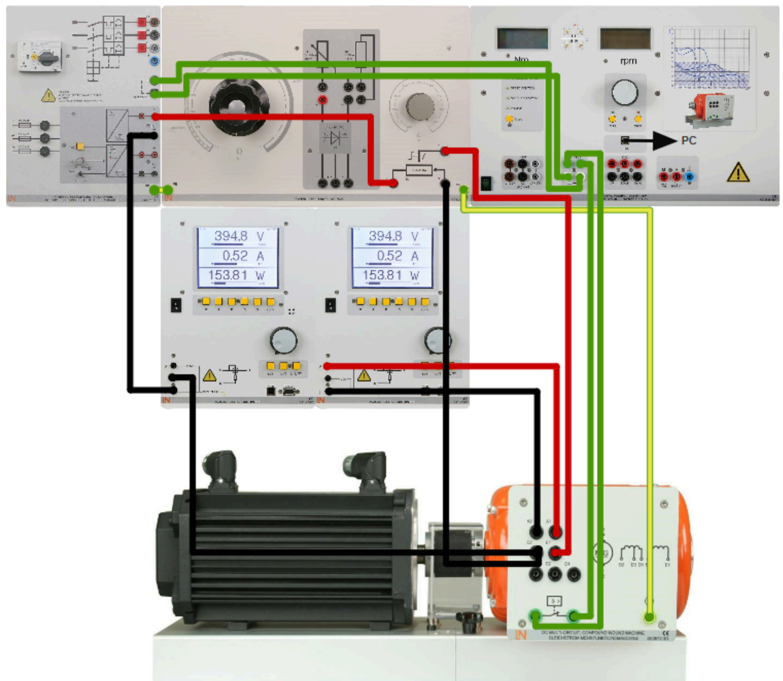
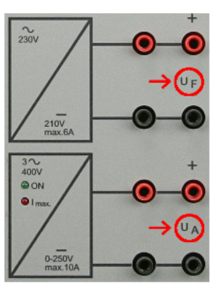


Figure 3.2.2 DC shunt-wound generators, separately excited wiring step by step

Record the characteristic of "V_G" as a function of "n" for different excitation currents

Required settings:

- Close the *ActiveServo* software in order to allow for manual operation.
- Brake mode: "Speed control"
- Field regulator: maximum value
- DC power supply unit:



	300 W
Armature circuit:	-
Excitation circuit:	210 V (from terminals U_F)

- Multimeter measurement mode: arithmetic mean

Figure 3.2.3 Require setting to obtain correct values

2. **Next set the generator into operation and drive the motor up to a speed of 2000 rpms. Set the field regulator to the excitor currents listed. Begin at 0 mA and continue until you record the generator voltage produced at different rpm.**
3. **Graph the generator voltage vs the different rpm**
4. **Answer some questions on the generator voltage and the excitor current.**

Part 2

5. **Flip the polarity of the generator and the rotation direction of the motor. Record the generator voltage before and after.**
6. **Answer a question on the effect that changing the polarity had on the generator voltage.**

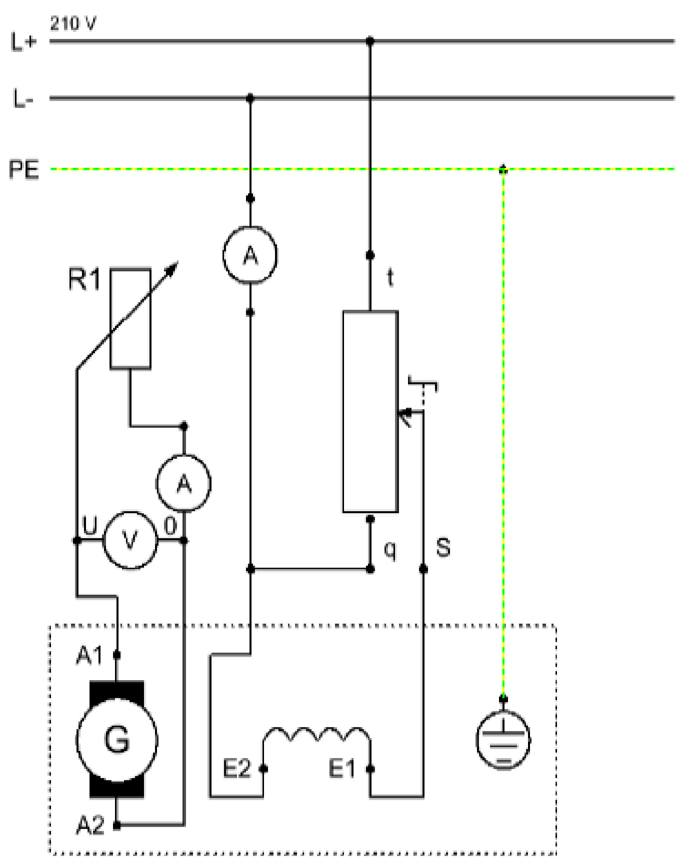
Experiment procedure:

- Before putting the machine into operation, set up the excitation winding in accordance with the nominal data by means of the field regulator. You may have to make adjustments as necessary.
- First run the drive motor up to a speed of 2000 rpm
- Measure the generator voltage V_G
- Now change the polarity of the exciter winding and then the rotation direction of the drive motor
- Measure the generator voltage V_G after each modification

Figure 3.2.4 Settings and polarity to measure the generator voltage

Part 3

7. **Next wire up the DC generator with the following set up for "Load Characteristics".**



Circuit diagram for shunt-wound generator, separately excited "Load Characteristics"

Figure 3.2.5 shunt wound generator, separately excited, "Load Characteristics"

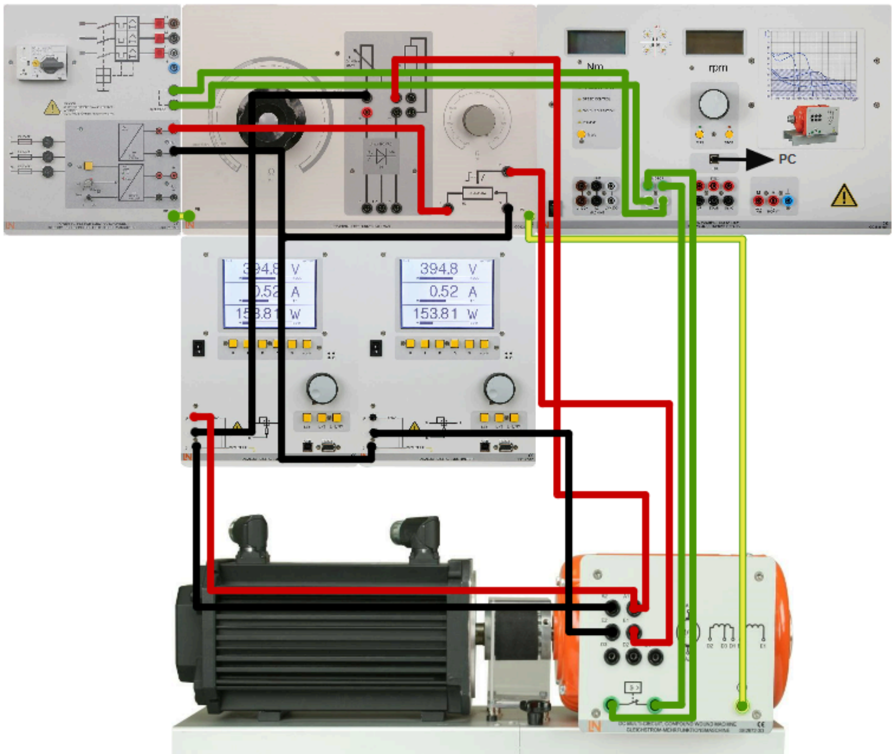
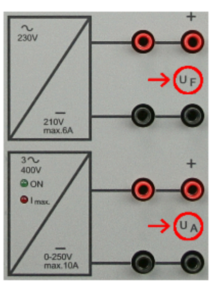


Figure 3.2.6 Shunt wound generator, separately excited, step by step wiring

Record the load characteristics of the generator with various exciter currents

Required settings:

- Brake mode: "Speed control"
- DC power supply unit:



	300 W
Armature circuit:	220 V (from terminals U_A)
Excitation circuit:	220 V (from terminals U_A)

Field regulator: minimum setting (0 Ω, corresponding to an excitation field at nominal voltage)
Load resistor: maximum (1 kΩ approx.)

Figure 3.2.7 Setting for DC Generator

8. **Drive motor is to be set at 2000 rpm. Record the armature currents specified in the table with two different nominal currents of 50% and 100%.**
9. **Measure the generator voltage and calculate the power output when different armature currents are utilized.**

10. **Graph both the power output and nominal current (blue). Also graph the generator voltage and the nominal current (red).**

4. ANALYSIS

4.1. Experimental Results

Standstill voltage = 60.4

Part 1

Experiment procedure:

- Before putting the machine into operation, set up the exciter winding in accordance with the nominal data by means of the field regulator. You may have to make adjustments as necessary.
- Put the generator into operation
- First run the drive motor up to a speed of 2000 rpm
- Use the field regulator to set the exciter currents specified in the table
- Begin at $I_{exc.}$= 0 mA. For this purpose you should short out the exciter winding via the field regulator (S - q).
- Measure the generator voltage V_G produced at each speed as you lower the speed step by step (see table)

	Iexc.=0mA	Iexc.=50mA	Iexc.=70mA	Iexc.=90mA
n/rpm	Vg/V	Vg/V	Vg/V	Vg/V
2000	6.65	180.00	189.80	189.80
1800	5.99	162.00	171.00	171.00
1600	5.30	146.20	152.20	152.20
1400	4.57	124.00	133.50	133.50
1200	3.80	108.50	114.80	114.80

Table Chart

Table 4.1.1(a) Generator voltage V_G record

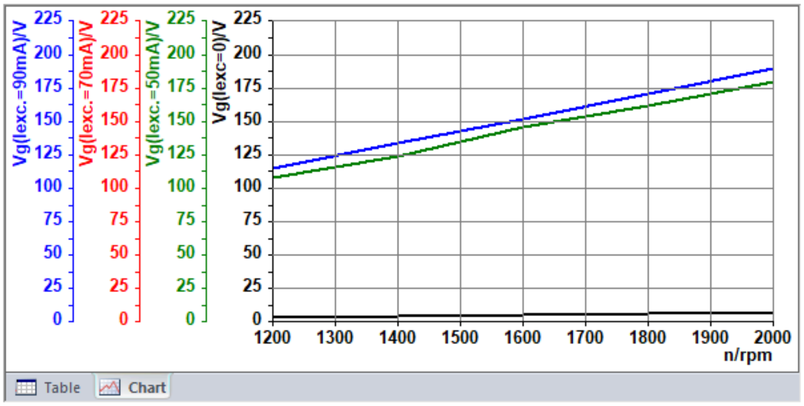
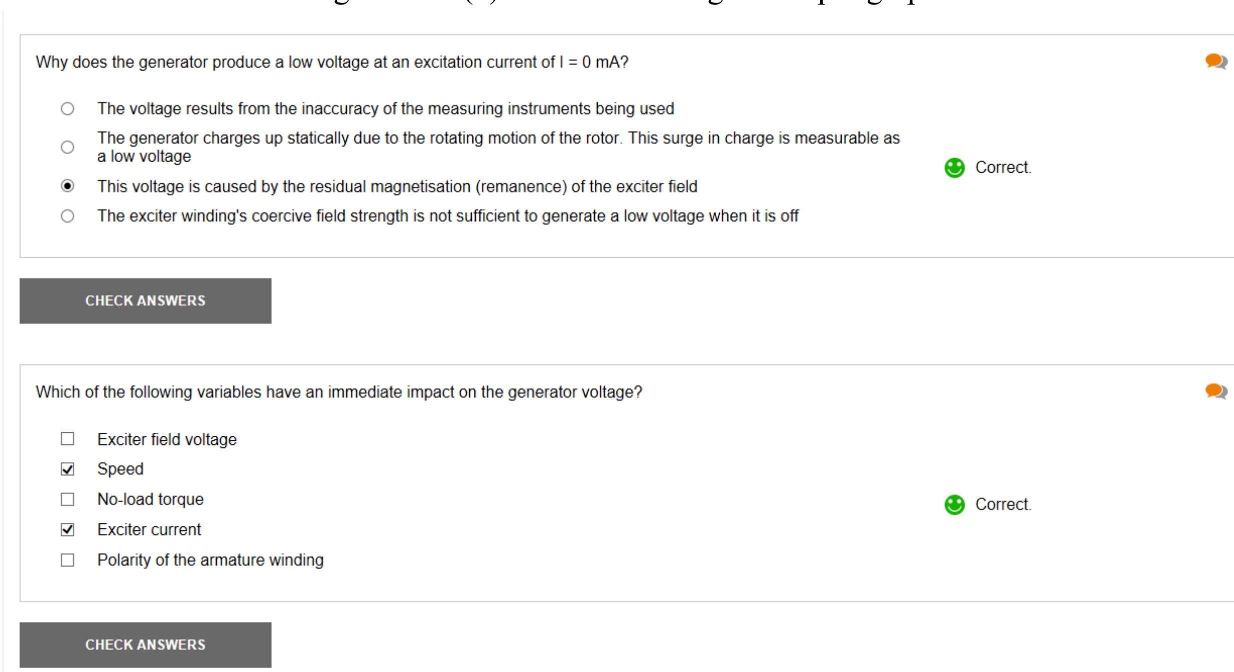


Figure 4.1.1(b) Generator voltage vs n/rpm graph



Why does the generator produce a low voltage at an excitation current of I = 0 mA?

- ○ The voltage results from the inaccuracy of the measuring instruments being used
- ○ The generator charges up statically due to the rotating motion of the rotor. This surge in charge is measurable as a low voltage
- ◉ This voltage is caused by the residual magnetisation (remanence) of the exciter field
- ○ The exciter winding's coercive field strength is not sufficient to generate a low voltage when it is off

Correct.

CHECK ANSWERS

Which of the following variables have an immediate impact on the generator voltage?

- ☐ Exciter field voltage
- ☑ Speed
- ☐ No-load torque
- ☑ Exciter current
- ☐ Polarity of the armature winding

Correct.

CHECK ANSWERS

Figure 4.1.2 Questions on generator voltage and excitation current

Part 2 - Properties of generator when polarity of exciter winding is reversed

Recorded generator voltage below:

VG = 187.5 V (before reversing polarity)

VG = -187.1 V (after reversing polarity)

Which of the following statements is true?

- ○ The polarity of the generator voltage is independent of the generator's rotation direction
- ● The polarity of the exciter winding and the rotation direction of the generator determine the polarity of the generator voltage — Correct.
- ○ The polarity of the generator voltage cannot be changed as it is fixed to the same polarity by the manufacturer

CHECK ANSWERS

Figure 4.1.3 Questions on the effect of change in polarity of generator voltage

Part 3 - Load characteristics

The electrical power output is computed as follows:

$$P_2=V_G*I_G,\ V_G[V],\ I_G[A],\ P_2[W]$$

100% of nominal exciter current

Figure 4.1.4 Formula utilized to calculate power output

Nominal exciter current at 100% = 0.06A

100% of nominal exciter current

Ig/A	0.3	0.6	0.8	1.0	1.2	1.4
Vg/V	173.7	161.4	149.5	136.4	124.2	110.0
P2/W	62.0	97.1	120.3	135.6	148.7	154.5

Table Chart

Table 4.1.2 power output and voltage generator for 100% of nominal current

100% of nominal exciter current

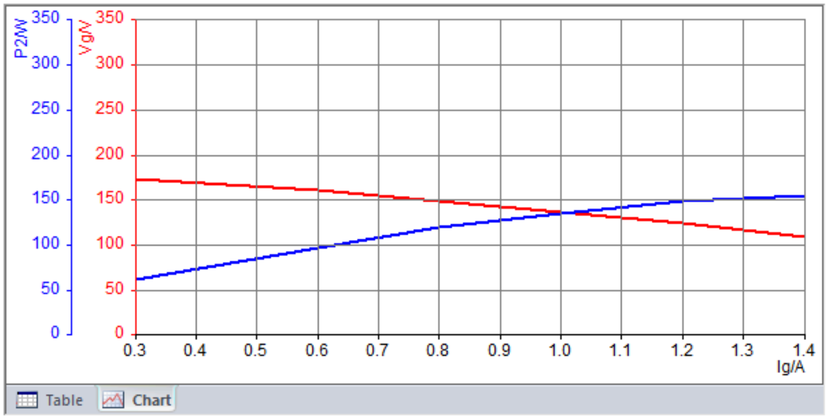


Figure 4.1.5 Graph of power output (p2/w) vs Ig/A for 100% of nominal current (blue) and generator voltage vs Ig/A (red)

Nominal exciter current at 50% = 0.03A

50% of nominal exciter current

Ig/A	0.3	0.6	0.8	1.0	1.2	1.4
Vg/V	151.0	137.7	122.3	107.7	95.5	79.9
P2/W	54.8	80.7	97.4	110.2	114.2	112.6

Table Chart

Table 4.1.3 generator voltage and power output for 50% of nominal current

50% of nominal exciter current

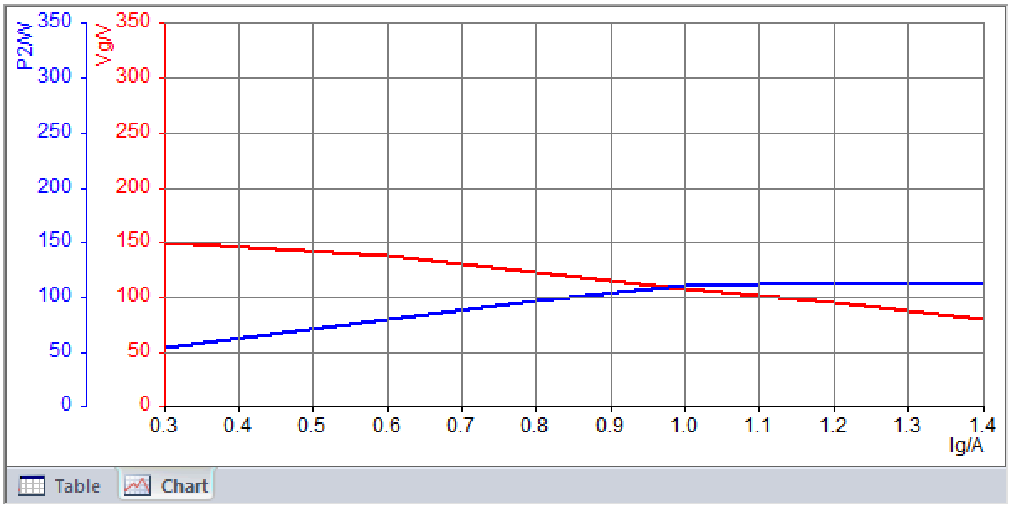


Figure 4.1.6: Graph of power output (P2w) vs. Ig/A for 50% of nominal current (blue), and generator voltage vs Ig/A in (red)

4.2. Data Analysis

Part 1:

Table 4.1a shows the generator voltage (V_G) as a function of rotational speed (n) at various excitation currents (I_exc). As expected, there is a clear trend showing that the generator voltage increases with both the rotational speed and the excitation current. This is a classic behavior of DC generators where the induced voltage in a conductor within a magnetic field is proportional to the speed of the conductor and the strength of the magnetic field.

Figure 4.1b represents the data from the table and visually confirms this relationship. The lines representing higher excitation currents are consistently above those with lower excitation currents, demonstrating higher voltages for any given speed. This illustrates the principle that greater excitation current increases the magnetic field strength in the generator, which in turn increases the generated voltage.
Figure 4.1.2 highlights that speed and exciter current have an immediate impact on generator voltage, confirming the direct correlation observed in the data. Speed affects the rate at which the magnetic field cuts the conductors, and exciter current affects the strength of the magnetic field, both key factors in the induced voltage.

Part 2:

VG = 187.5 V (before reversing polarity)
VG = -187.1 V (after reversing polarity)
The data indicates that reversing the polarity of the exciter winding resulted in the generator voltage also reversing its polarity, which is consistent with the principles of electromagnetic

induction. The change in polarity of the exciter winding changes the direction of the magnetic field within the generator.
Figure 4.1.3 further clarifies that "The polarity of the exciter winding and the rotation direction of the generator determine the polarity of the generator voltage." This statement is indeed correct and is supported by the data collected.

Part 3:
Looking Table 4.1.3 and Figure 4.1.5 (100% of nominal exciter current (0.06A)), we can see the following trends:

- As the armature current increases, the generator voltage decreases.
- The power output initially increases with the armature current, reaches a peak, and then begins to decrease.
- This suggests that at higher loads (higher armature currents), the internal voltage drop in the generator increases due to the armature reaction and ohmic losses in the windings, resulting in a lower terminal voltage. However, the power output increases up to a certain point because the product of voltage and current (P = V × I) increases as long as the decrease in voltage isn't too steep.
- The peak power output at 100% excitation is observed at 1.2A of armature current, where the power output is approximately 148.7 watts. Beyond this point, further increases in current lead to a decrease in power output, likely due to the excessive voltage drop overcoming the increase in current.

Table 4.1.3 shows and Figure 4.1.6 trends are similar, with the following observations:

- The generator voltage decreases with increasing armature current, albeit starting from a lower initial voltage than in the 100% excitation scenario due to the lower excitation current.
- The power output increases with the armature current, peaks, and then starts to decrease.
- At 50% excitation, the peak power output occurs at an armature current of 1.2A, with a power output of approximately 114.2 watts, which is lower than the peak observed at 100% excitation.This aligns with the expectation that a lower excitation current will result in a lower magnetic field strength and, consequently, a lower induced voltage and power output.

5. CONCLUSIONS

In conclusion this lab presented an in-depth exploration of the characteristics and performance of DC Shunt-wound generators through a series of meticulously designed experiments. The primary objectives were to understand the behavior of generator voltage under varying rpm, the impact of excitor current changes, and the effects of altering the polarity and load on the generator's output.

The first part of the experiment successfully demonstrated the relationship between the generator voltage and rpm, with data indicating a consistent pattern as excitor currents were adjusted. This finding underscores the predictable nature of generator output under controlled speed and excitation conditions, highlighting the importance of precise current regulation in managing generator performance.

In the second part, reversing the polarity of the exciter winding produced a significant effect on the generator voltage, effectively inverting its output. This result not only confirms the theoretical predictions about the influence of magnetic field direction on generated voltage but also provides a practical understanding of how generator output can be manipulated through simple adjustments in wiring and orientation.

The third part focused on analyzing the generator's load characteristics by examining how different armature currents, representing varying load conditions, affect the power output and voltage of the generator. The experiments revealed a direct correlation between load intensity and generator output, with power and voltage graphs illustrating the generator's response to changes in demand. These findings are crucial for optimizing generator performance and efficiency in real-world applications, where varying loads can significantly impact electrical output.

6. References

Dr. Woonki Na, Lab 5

Lab Report Evaluation Rubric

Course: ECE _______ Date: _____________

Evaluate each component using a weighted scale based upon the following criteria:

FORMATTING RUBRIC

	Poor	Excellent	Weight	Score
Title Page and Table of Contents	Missing or does not Follow Standards	Fully Follows Guidelines and Standards	5	
General Format of Technical Paper	Does not Follow Guidelines and Standards	Fully Follows Guidelines and Standards	5	
Overall Structure of Technical Paper	Missing Sections or Information in Incorrect Locations	Appropriate Sections; Information in Correct Location	10	

WRITING RUBRIC

	Poor	Excellent	Weight	Score
Spelling and Grammar	Many Errors	Minor or No Errors	10	
Sentence Structure and Transitions	Poor Structure	Well Structured	10	
Focus and Organization	Unorganized and Lacks Clarity; Poor Presentation	Well Organized and Reads Clearly; Good Presentation	10	

TECHNICAL CONTENT RUBRIC

	Poor	Excellent	Possible Points	Score
Objectives and Theoretical Background	Meaningless Objectives; Terms Unrelated to Content	Objectives Clearly Stated; Theory Well Presented	10	
Experimental Procedure	Experimental Procedure Unclear and Incomplete	Experimental Procedure Valid and Complete	20	
Analysis and Technical Conclusions	Inconsistent with Observations or Objectives	Consistent with Observations and Objectives	20	

Overall average score __________